Excel Built-In Volatile Functions (v2)

In Microsoft Excel, **volatile functions** are functions that recalculate every single time Excel calculates a worksheet, regardless of whether the cells they depend on have changed. Because they trigger continuous recalculations, overusing them can significantly slow down workbook performance.

Conversely, standard lookup functions like XLOOKUP, VLOOKUP, and MATCH are **not** volatile; they only recalculate when their specific source data changes.

Below is the updated, comprehensive list of the core built-in volatile functions available in Excel, including modern dynamic array functions:

Function Name	Classification	Description	Common Use Case / Example
RAND()	Fully Volatile	Returns a random decimal number greater than or equal to 0 and less than 1.	Generating dummy data or random percentages.
RANDBETWEEN()	Fully Volatile	Returns a random integer number between the numbers you specify.	=RANDBETWEEN(1, 100) to assign a random ID.
RANDARRAY()	Fully Volatile (Modern)	Returns an array of random numbers between 0 and 1, or integers between specified bounds. Available in Excel 365/2021+.	=RANDARRAY(5, 2, 1, 10, TRUE) to fill a 5x2 grid with integers.
NOW()	Fully Volatile	Returns the current date and time.	Timestamping or finding the exact elapsed time.
TODAY()	Fully Volatile	Returns the current date.	=TODAY() - A2 to calculate aging days past due.
OFFSET()	Fully Volatile	Returns a reference to a range that is a specified number of rows and columns from a cell or range of	Creating dynamic named ranges (though INDEX is preferred for performance).

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INDIRECT()	Fully Volatile	Returns the reference specified by a text string.	Dynamically pulling data from different sheets based on a cell value.
INFO()	Fully Volatile	Returns information about the current operating environment.	=INFO("osversion") to check the operating system.
CELL()	Conditionally Volatile	Returns information about the formatting, location, or contents of a cell. *Volatile only when certain arguments (like "filename") are used, or when the second argument is omitted.*	=CELL("filename") to extract the current file path.
SUMIF() / COUNTIF()	Open-Volatile Only	Standard conditional evaluation functions. *They are not volatile during editing, but will force a recalculation once when the workbook is opened.*	=SUMIF(A:A, "Sales", B:B)

VBA Custom Functions

User-Defined Functions (UDFs) written in VBA can be explicitly turned into volatile functions by adding the statement Application.Volatile inside the macro code. This is useful when the custom macro needs to track external spreadsheet environment changes.

Best Practices to Manage Volatility

- **Minimize Usage:** Use non-volatile alternatives where possible. For instance, swap OFFSET for a non-volatile INDEX reference.

- **Convert to Values:** If you are using random-generation functions to populate data, copy the results and paste them as values once generated to stop continuous recalculation.
- **Avoid Volatile Arguments:** Passing a volatile function inside an efficient function (e.g., =XLOOKUP(TODAY(), A:A, B:B)) forces the entire formula to recalculate as if it were volatile.
- **Manual Calculation Mode:** For massive workbooks using these functions, consider switching Excel's calculation options from "Automatic" to "Manual" (found under the Formulas tab).